

## Assignment No 08

**AIM:** Create a multi-threaded Java application that simulates a basic chat system. Each user (thread) sends and receives messages. Use `isAlive()` to check the status of threads and `join()` to ensure proper synchronization. Implement thread priorities to handle high-priority messages and demonstrate thread suspension, resumption, and stopping.

### Code:

```
class ChatUser extends Thread {

    private String userName;
    private boolean suspended = false;
    private boolean stopped = false;

    public ChatUser(String userName) {
        this.userName = userName;
    }

    public void run() {
        try {
            for (int i = 1; i <= 5; i++) {

                synchronized (this) {
                    while (suspended) {
                        wait();
                    }
                }

                if (stopped)
                    break;

                System.out.println(userName + " sends message " + i);
                Thread.sleep(1000);
            }
        } catch (InterruptedException e) {
            System.out.println(userName + " interrupted.");
        }

        System.out.println(userName + " has left the chat.");
    }

    public synchronized void suspendUser() {
        suspended = true;
    }
}
```

```

    public synchronized void resumeUser() {
        suspended = false;
        notify();
    }

    public void stopUser() {
        stopped = true;
    }
}

public class MultiThreadChat {

    public static void main(String[] args) {

        ChatUser user1 = new ChatUser("Siddharth");
        ChatUser user2 = new ChatUser("Rutik");
        ChatUser user3 = new ChatUser("Tejas");

        user3.setPriority(Thread.MAX_PRIORITY);
        user1.setPriority(Thread.NORM_PRIORITY);
        user2.setPriority(Thread.MIN_PRIORITY);

        user1.start();
        user2.start();
        user3.start();

        System.out.println("Is Siddharth alive? " + user1.isAlive());
        System.out.println("Is Rutik alive? " + user2.isAlive());
        System.out.println("Is Tejas alive? " + user3.isAlive());

        try {
            Thread.sleep(3000);

            System.out.println("Suspending Rutik...");
            user2.suspendUser();

            Thread.sleep(3000);

            System.out.println("Resuming Siddharth...");
            user2.resumeUser();

            Thread.sleep(3000);

            System.out.println("Stopping Tejas...");
            user1.stopUser();

            user1.join();

```

```

        user2.join();
        user3.join();

    } catch (InterruptedException e) {
        e.printStackTrace();
    }

    System.out.println("All users have finished chatting.");
}
}

```

## Output:

```

"C:\Program Files\Java\jdk-21\bin\java.exe" "-java
Is Siddharth alive? true
Is Rutik alive? true
Is Tejas alive? true
Rutik sends message 1
Tejas sends message 1
Siddharth sends message 1
Rutik sends message 2
Siddharth sends message 2
Tejas sends message 2
Siddharth sends message 3
Tejas sends message 3
Rutik sends message 3
Suspending Rutik...
Siddharth sends message 4
Tejas sends message 4
Siddharth sends message 5
Tejas sends message 5
Tejas has left the chat.
Siddharth has left the chat.
Resuming Siddharth...
Rutik sends message 4
Rutik sends message 5
Rutik has left the chat.
Stopping Tejas...
All users have finished chatting.

Process finished with exit code 0

```

